

PHYSICS CLASS 9

THEORY GUESS PAPER

FBISE SSC-I | 2nd Annual 2026 | Final revision for 8 October 2026

Fakhar STEM Sphere | 42 short questions + 12 numericals + 12 long questions

Scope: Only SLOs labelled Summative for the theory examination in the supplied framework. No MCQs, formative-only questions or summative-for-PBA questions.

Priority: P1 = practise first: topic or question style appears in the supplied sample papers. P2 = complementary theory coverage. These are revision priorities; actual appearance probabilities cannot be established from two papers.

Instructions: Answer in a separate notebook. Give reasons, labelled diagrams and SI units where required. Short questions: suggested 3 marks each. Numericals and long questions: suggested 6 marks each. These are practice allocations, not an official board marking scheme.

A. SHORT QUESTIONS | S1-S10

Write concise, relevant answers. For comparisons, give clear paired points.

S1. Differentiate between base and derived physical quantities. Give one example of each and write its SI unit.

P1 | Theory SLOs: P-09-A-03, P-09-A-04

S2. Give three differences between scalar and vector quantities. Explain why displacement is a vector.

P1 | Theory SLOs: P-09-A-07, P-09-A-09

S3. Explain scientific notation with an example. Show that the prefix micro is one thousand times smaller than milli.

P1 | Theory SLOs: P-09-A-05, P-09-A-06

S4. A balance gives slightly different readings for the same object on repeated measurements. Identify the type of error and suggest two ways to reduce it.

P1 | Theory SLOs: P-09-A-17

S5. Differentiate between accuracy and precision. Can readings be precise but inaccurate? Give an example.

P1 | Theory SLOs: P-09-A-18

S6. Differentiate between physical and non-physical quantities with examples. Why must a measurement include both magnitude and unit?

P2 | Theory SLOs: P-09-A-01, P-09-A-02

S7. Distinguish between translatory, rotatory and vibratory motion. Give one example of each.

P2 | Theory SLOs: P-09-B-01

S8. Differentiate between speed and velocity, and between average and instantaneous speed.

P1 | Theory SLOs: P-09-B-02, P-09-B-05

S9. Define acceleration and derive its SI unit. Distinguish between uniform velocity and uniform acceleration.

P2 | Theory SLOs: P-09-B-06, P-09-B-07, P-09-B-08

S10. Sketch distance-time graphs for a body at rest, moving at constant speed, and decelerating. Explain what the gradient shows.

P1 | Theory SLOs: P-09-B-09, P-09-B-11

A. SHORT QUESTIONS | S11-S21

Suggested 3 marks per question | Continue in your answer notebook.

S11. What do the gradient and the area under a speed-time graph represent? Explain using their units.

P1 | Theory SLOs: P-09-B-12, P-09-B-14

S12. State Newton's first law. Use inertia to explain why passengers move forward when a moving bus stops suddenly.

P1 | Theory SLOs: P-09-B-18, P-09-B-27

S13. Differentiate between mass and weight. Define gravitational field strength and write its SI unit.

P2 | Theory SLOs: P-09-B-19, P-09-B-20

S14. Name the three fundamental interactions listed in the supplied framework: gravitational, strong and electroweak. State one feature of each and name the Pakistani Nobel laureate associated with electroweak unification.

P1 | Theory SLOs: P-09-B-25

S15. Differentiate between contact and non-contact forces. Draw a labelled free-body diagram of a book resting on a horizontal table.

P2 | Theory SLOs: P-09-B-23, P-09-B-24, P-09-B-26

S16. State Newton's third law and give two examples. Why do action and reaction forces not cancel each other on a single body?

P1 | Theory SLOs: P-09-B-31, P-09-B-32

S17. State Newton's second law in terms of acceleration. Mention two situations in which Newtonian mechanics is not an adequate description.

P2 | Theory SLOs: P-09-B-30, P-09-B-33

S18. Explain how friction dissipates energy. Distinguish rolling from sliding friction and state one method of reducing friction.

P1 | Theory SLOs: P-09-B-35, P-09-B-37, P-09-B-38

S19. Explain how a falling object reaches terminal velocity. What are its resultant force and acceleration at terminal velocity?

P2 | Theory SLOs: P-09-B-36

S20. Define momentum and impulse with their SI units. Explain why increasing the stopping time reduces the average force in a collision.

P1 | Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-42

S21. State the principle of conservation of momentum. Explain the recoil of a gun using this principle.

P2 | Theory SLOs: P-09-B-41

A. SHORT QUESTIONS | S22-S32

Suggested 3 marks per question | Continue in your answer notebook.

S22. Differentiate between like and unlike parallel forces. Define the moment of a force and explain why a door handle is placed far from its hinges.

P2 | Theory SLOs: P-09-B-43, P-09-B-44

S23. State the two conditions for complete equilibrium of a rigid body. Distinguish stable, unstable and neutral equilibrium using examples.

P2 | Theory SLOs: P-09-B-34, P-09-B-45

S24. Define centre of gravity. Explain why a racing car is more stable when it has a low centre of gravity and a wide base.

P2 | Theory SLOs: P-09-B-47, P-09-B-49, P-09-B-50

S25. Define centripetal force and state its direction. Identify the force that keeps the Moon in orbit and the force that keeps a stone moving in a circle on a string.

P1 | Theory SLOs: P-09-B-53, P-09-B-54

S26. Define spring constant and give its SI unit. Explain how Hooke's law is used in a spring balance.

P2 | Theory SLOs: P-09-B-56, P-09-B-59

S27. Define work done and write its formula and SI unit. Explain why holding a stationary bag involves no mechanical work on the bag.

P2 | Theory SLOs: P-09-B-60, P-09-B-61

S28. State the law of conservation of energy. Explain why a machine cannot continuously supply useful energy without an energy input.

P1 | Theory SLOs: P-09-B-68, P-09-B-69

S29. Define power and efficiency. Write the formulas for power and percentage efficiency, identifying the quantities used.

P1 | Theory SLOs: P-09-B-73, P-09-B-74

S30. Differentiate renewable and non-renewable energy sources. Give two examples of each and one environmental disadvantage of fossil fuels.

P2 | Theory SLOs: P-09-B-70, P-09-B-72

S31. Define pressure with its SI unit. Explain why a sharp knife cuts better than a blunt knife and why broad tyres reduce sinking in soft ground.

P1 | Theory SLOs: P-09-B-77, P-09-B-78

S32. Explain how a drinking straw demonstrates atmospheric pressure. Would an ordinary open drinking straw work on the Moon? Justify your answer.

P1 | Theory SLOs: P-09-B-80, P-09-B-81

A. SHORT QUESTIONS | S33-S42

Suggested 3 marks per question | Continue in your answer notebook.

S33. Differentiate between a barometer and a manometer. How can a rapid fall in atmospheric pressure help indicate a change in weather?

P1 | Theory SLOs: P-09-B-82, P-09-B-83, P-09-B-85

S34. Explain why liquid pressure increases with depth. State Pascal's law and name one hydraulic application.

P1 | Theory SLOs: P-09-B-84, P-09-B-86

S35. Define density and give its formula and SI unit. Explain why equal volumes of two substances can have different masses.

P1 | Theory SLOs: P-09-C-01

S36. Explain how particle motion changes as temperature rises. Relate this change to the internal energy of a substance.

P1 | Theory SLOs: P-09-C-03, P-09-C-05, P-09-C-06

S37. What is plasma? Give two examples. State the approximate Celsius temperature of absolute zero and explain its meaning.

P2 | Theory SLOs: P-09-C-04, P-09-C-05

S38. What is a thermometric property? Why are fixed points needed for thermometer calibration? Explain how a narrow capillary affects sensitivity.

P1 | Theory SLOs: P-09-C-07, P-09-C-08, P-09-C-11

S39. Define induced magnetism. Give two differences between a permanent magnet and an electromagnet and one use of each.

P1 | Theory SLOs: P-09-E-02, P-09-E-04, P-09-E-10

S40. Draw the magnetic field around a bar magnet, showing direction. Explain how the spacing of field lines indicates field strength.

P1 | Theory SLOs: P-09-E-05, P-09-E-07, P-09-E-08, P-09-E-09

S41. Compare the responses of paramagnetic and diamagnetic materials to an external magnetic field. Explain why soft iron is used for magnetic shielding.

P1 | Theory SLOs: P-09-E-12, P-09-E-15

S42. Briefly describe astronomy, particle physics and geophysics. Explain what is meant by a scientific theory and how evidence is used to develop it.

P1 | Theory SLOs: P-09-G-02, P-09-G-06

B. NUMERICAL PROBLEMS | N1-N6

Suggested 6 marks per question | Show data, formula, substitution, calculation and unit. Use $g = 9.8 \text{ m/s}^2$ unless otherwise stated. All data are provided for practice.

N1. (a) Convert 25 micrometres to kilometres. (b) Express 0.0000034 s using a suitable SI prefix. (c) Write 72,000,000 N in scientific notation.

P1 | Theory SLOs: P-09-A-05, P-09-A-06

N2. Two perpendicular displacements are 6 m east and 8 m north. Calculate the magnitude of their resultant and its angle measured north of east. Draw a labelled vector diagram.

P2 | Theory SLOs: P-09-A-10

N3. A car travels 6 m in 2 s, then 8 m in 4 s, then 10 m in 2 s. Calculate (a) total distance, (b) total time and (c) average speed. Explain why the mean of the three individual speeds is not appropriate here.

P1 | Theory SLOs: P-09-B-03, P-09-B-04

N4. A car accelerates uniformly from rest to 20 m/s in 5 s, continues at 20 m/s for 10 s, and then slows uniformly to rest in 5 s. Draw its speed-time graph. Calculate (a) initial acceleration, (b) acceleration while stopping and (c) total distance from the graph.

P1 | Theory SLOs: P-09-B-07, P-09-B-12, P-09-B-14

N5. A stone is dropped from rest and reaches the ground at 29.4 m/s. Ignore air resistance. Calculate (a) time of fall and (b) height of release, using the area under its speed-time graph.

P1 | Theory SLOs: P-09-B-10, P-09-B-12

N6. A 2.0 kg object changes velocity from 3.0 m/s to 7.0 m/s in the same direction in 0.20 s. Calculate (a) change in momentum, (b) impulse and (c) average resultant force.

P1 | Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-42

B. NUMERICAL PROBLEMS | N7-N12

Suggested 6 marks per question | Show data, formula, substitution, calculation and unit. Use $g = 9.8 \text{ m/s}^2$ unless otherwise stated. All data are provided for practice.

N7. A 2.0 kg trolley moving at 4.0 m/s strikes a stationary 3.0 kg trolley. They stick together. Neglect external horizontal forces. Calculate their common velocity and check that total momentum is conserved.

P2 | Theory SLOs: P-09-B-41

N8. A light horizontal beam is supported at its centre. A 30 N downward force acts 0.40 m to the left of the pivot. What downward force applied 0.60 m to the right balances the beam? Calculate each moment and the upward support force.

P2 | Theory SLOs: P-09-B-44, P-09-B-45

N9. An ideal spring extends by 4.0 cm under a force of 12 N. Calculate (a) its spring constant and (b) the force required for a 7.0 cm extension, assuming force remains proportional to extension.

P2 | Theory SLOs: P-09-B-56

N10. A football of mass 0.45 kg is kicked vertically upward at 22 m/s. Ignore air resistance in part (a). Calculate (a) its maximum height using energy conservation. If it actually rises to only 20 m, calculate (b) work done against air resistance on the upward journey.

P1 | Theory SLOs: P-09-B-61, P-09-B-64, P-09-B-65, P-09-B-66

N11. A hydraulic lift supports a car of mass 2500 kg. The small and large piston areas are 6.0 cm^2 and 90 cm^2 . Neglect piston weight and friction. Calculate (a) the car's weight, (b) the required force on the small piston and (c) the fluid pressure.

P1 | Theory SLOs: P-09-B-19, P-09-B-86

N12. A satellite moves in a circular orbit of radius $7.0 \times 10^6 \text{ m}$ and period 6000 s. Calculate (a) the distance travelled in one orbit and (b) its average orbital speed. Convert the speed to km/s. Use $\pi = 3.142$.

P1 | Theory SLOs: P-09-F-01

C. LONG QUESTIONS | L1-L4

Suggested 6 marks per question | Attempt every part. Draw neat labelled diagrams where asked.

L1. Motion graphs

- (a) Sketch a distance-time graph for an object initially at rest, then moving at constant speed, then accelerating and finally at rest again. (3 marks)
- (b) Explain how the gradient shows speed. (1 mark)
- (c) Using a simple speed-time graph, show that its gradient gives acceleration and its area gives distance travelled. (2 marks)

P1 | Theory SLOs: P-09-B-09, P-09-B-11, P-09-B-12, P-09-B-15

L2. Newton's laws in daily life

- (a) State Newton's three laws of motion. (3 marks)
- (b) Apply them respectively to a passenger in a suddenly stopping bus, a trolley pushed by an unbalanced force, and a swimmer moving forward. Identify the two bodies in the action-reaction pair. (3 marks)

P2 | Theory SLOs: P-09-B-27, P-09-B-30, P-09-B-31, P-09-B-32

L3. Friction and terminal velocity

- (a) Explain how friction in machine parts transfers mechanical energy to internal energy. (2 marks)
- (b) Explain two methods of reducing friction. (2 marks)
- (c) Describe the changing forces and acceleration on a falling object until it reaches terminal velocity. (2 marks)

P1 | Theory SLOs: P-09-B-35, P-09-B-36, P-09-B-38

L4. Momentum, impulse and safety

- (a) Define momentum and impulse and state conservation of momentum. (3 marks)
- (b) Show that average resultant force equals change in momentum divided by time. (1 mark)
- (c) Explain how airbags and a cricketer moving their hands backward while catching a ball reduce the average force. (2 marks)

P1 | Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-41, P-09-B-42

C. LONG QUESTIONS | L5-L8

Suggested 6 marks per question | Attempt every part. Draw neat labelled diagrams where asked.

L5. Moments and stability

- (a) Define the moment of a force and state the principle of moments. (2 marks)
- (b) Draw a labelled balanced-beam diagram and explain its rotational equilibrium. (2 marks)
- (c) Explain how a low centre of gravity and a broad base improve stability, using a vehicle as an example. (2 marks)

P2 | Theory SLOs: P-09-B-34, P-09-B-44, P-09-B-45, P-09-B-49, P-09-B-50

L6. Circular motion and orbital speed

- (a) Define centripetal force and show its direction on a circular-path diagram. (2 marks)
- (b) Identify its source for a stone on a string and the Moon orbiting Earth. (2 marks)
- (c) Derive average orbital speed $v = 2\pi r / T$ from the distance travelled in one circular orbit. (2 marks)

P1 | Theory SLOs: P-09-B-53, P-09-B-54, P-09-F-01

L7. Kinetic and gravitational potential energy

- (a) Derive kinetic energy $= (1/2)mv^2$ using the triangular area under a speed-time graph for uniform acceleration from rest, together with $F = ma$ and $W = Fs$. (3 marks)
- (b) Derive gravitational potential energy $= mgh$ for lifting an object at constant speed. (2 marks)
- (c) State the energy conversion during a fall without air resistance. (1 mark)

P1 | Theory SLOs: P-09-B-61, P-09-B-64, P-09-B-65, P-09-B-66

L8. Energy resources, power and efficiency

- (a) Describe the energy transfers in a hydroelectric power station, identifying the turbine and generator. (2 marks)
- (b) Compare solar and fossil-fuel electricity generation in terms of availability and environmental effects. (2 marks)
- (c) Define power and percentage efficiency and write their formulas. (2 marks)

P2 | Theory SLOs: P-09-B-68, P-09-B-71, P-09-B-72, P-09-B-73, P-09-B-74

C. LONG QUESTIONS | L9-L12

Suggested 6 marks per question | Attempt every part. Draw neat labelled diagrams where asked.

L9. Liquid pressure and hydraulic machines

- (a) Derive the gauge pressure at depth h in a liquid: $p = \rho gh$, using a liquid column of cross-sectional area A . (3 marks)
- (b) State Pascal's law. Draw a labelled hydraulic lift and explain how it multiplies force, using $F_1/A_1 = F_2/A_2$. (3 marks)

P1 | Theory SLOs: P-09-B-84, P-09-B-86

L10. Thermometers

- (a) Explain the construction and working of a liquid-in-glass thermometer and a thermocouple thermometer. (4 marks)
- (b) Explain how capillary width and bulb size affect the sensitivity of a liquid-in-glass thermometer, and how a fixed capillary length links sensitivity to range. (2 marks)

P1 | Theory SLOs: P-09-C-07, P-09-C-08, P-09-C-09, P-09-C-10, P-09-C-11

L11. Magnetization and magnetic domains

- (a) Explain magnetization and demagnetization using domain theory. (2 marks)
- (b) Describe one method of magnetizing a steel bar and one method of demagnetizing it. (2 marks)
- (c) Explain induced magnetism in soft iron and draw the field pattern around a bar magnet with arrows. (2 marks)

P1 | Theory SLOs: P-09-E-02, P-09-E-04, P-09-E-07, P-09-E-11

L12. Applications of magnets

- (a) Explain how an electromagnet operates an electric bell, including the repeated making and breaking of the circuit. (2 marks)
- (b) Explain how magnetic recording stores information in a magnetizable medium. (2 marks)
- (c) Explain magnetic shielding with soft iron and why magnetic storage media should be protected from strong external magnetic fields. (2 marks)

P1 | Theory SLOs: P-09-E-10, P-09-E-14, P-09-E-15

Teacher scope check: Excluded PBA SLOs A-12 to A-16, A-19, B-21, B-22, B-48, B-57 and B-58, plus all Domain H formative/PBA content. Accuracy/errors (A-17/A-18), field diagrams (E-07) and thermometer theory (C-07 to C-11) remain eligible theory topics.

Sources used: Supplied SSC-I summative assessment framework; supplied Physics 2nd Annual 2025 subjective sample; supplied Physics 1st Annual 2026 D subjective sample. The attachment set contains a framework and two papers, not a complete textbook. The framework controls eligibility where a sample includes a broader demand.